Individual Assignment 5

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1. Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```r
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")


# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))


# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
```

```r
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")


# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
```

```r
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~  "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
```

```
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()

# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()
```

2. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: log_ave_lit
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangle your data

```r
#creating new object frm aq_ins_clean
ostracods <- aq_ins_clean |>
  #filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#display 10 random observations
ostracods |>
  slice_sample(n = 10)
```

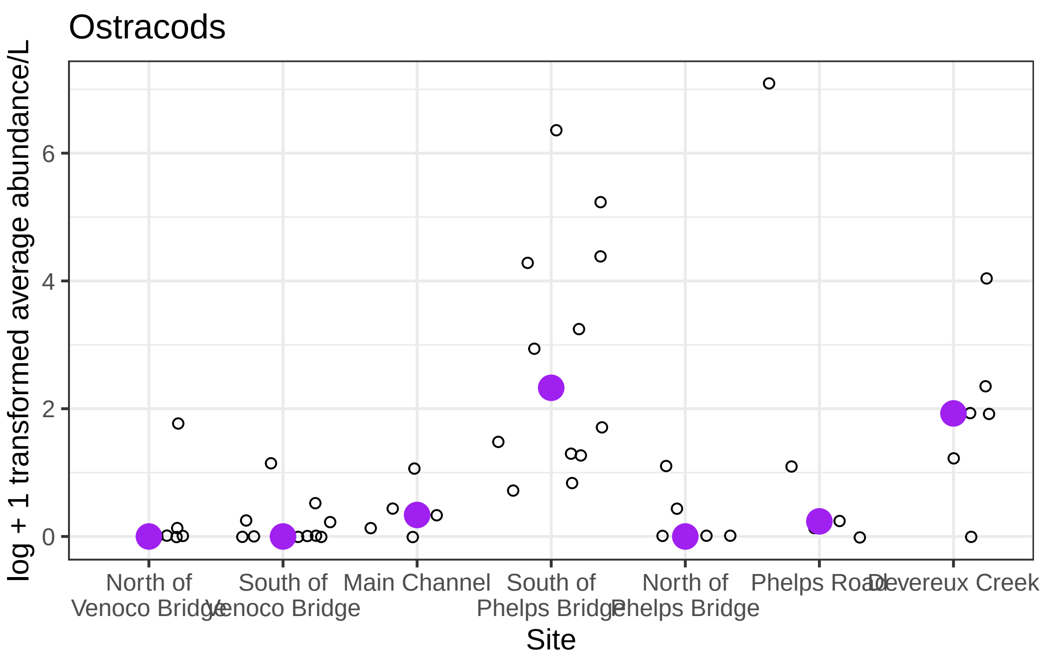
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-02-21_NEC  2023-02-21 00:00:00 NEC   ostr~   0       8.56   NA       8.8
 2 2022-01-28_NDC  2022-01-28 00:00:00 NDC   ostr~   9.47    7.65    2.00    9.55
 3 2022-08-10_NPB  2022-08-10 00:00:00 NPB   ostr~   2.67    8.28    8.84    9.81
 4 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  ostr~   0      NA      NA      NA
 5 2021-04-15_NPB  2021-04-15 00:00:00 NPB   ostr~ 185.      8.09    3.57    8.24
 6 2023-08-12_NEC  2023-08-12 00:00:00 NEC   ostr~   0      NA      NA      NA
 7 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA      47      14.0
 8 2023-01-28_NPB1 2023-01-28 00:00:00 NPB1  ostr~   0       7.66   NA       9.74
 9 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR  ostr~   0       8.86   45.2    10.2
10 2024-01-28_NPB  2024-01-28 00:00:00 NPB   ostr~   3.33    7.43    3.08    6.6
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```r
# base layer
ostracods_plot <- ggplot(data = ostracods,
                   # x-axis: site
                   # y-axis: log + 1 transformed average abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
```

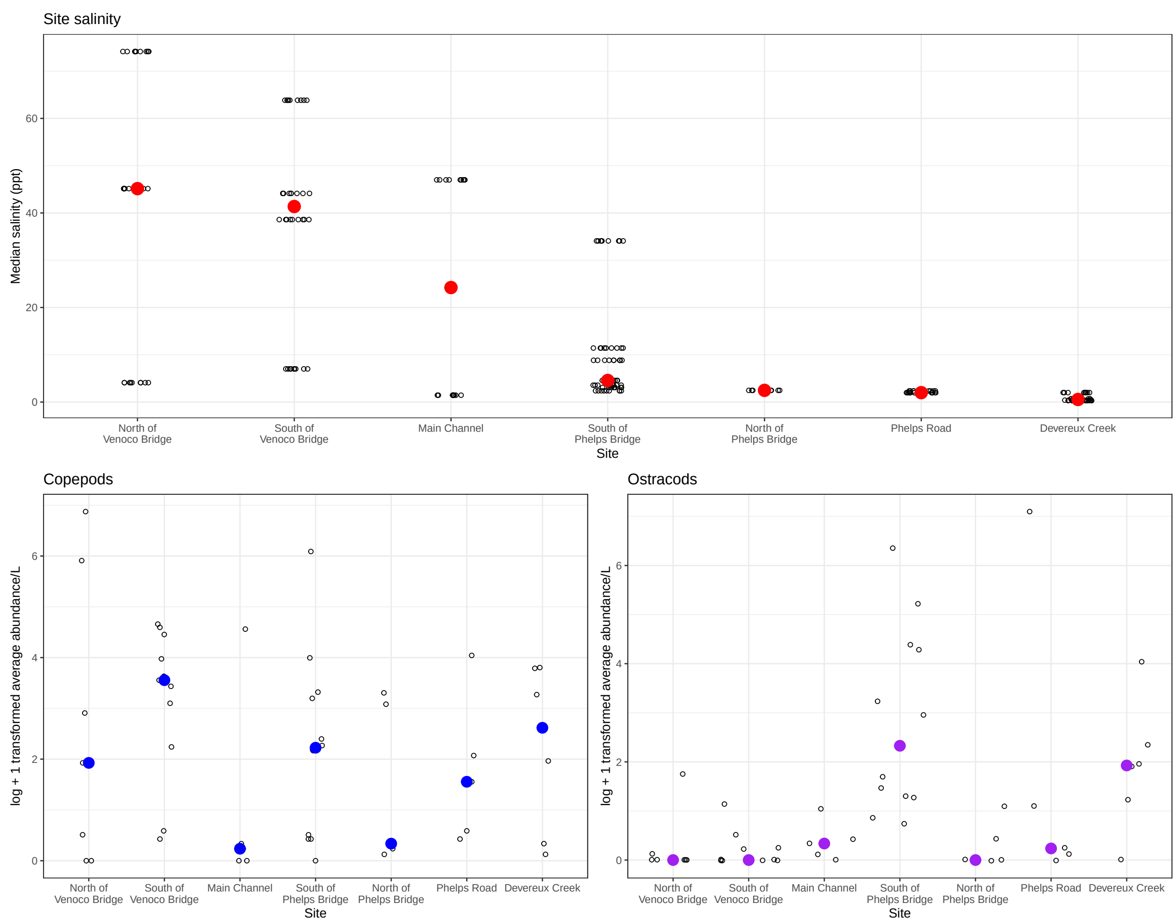
```
  # first layer: points
  geom_jitter(shape = 21) +
  # second layer: median points
  # changing size, color, shape
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "purple") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # custom theme
  theme_bw()

ostracods_plot
```



d. Create a multi panel figure

```
#using patchwork to create multipanel figure
# *used chatgpt for help*
salinity /
  (copepods | ostracods_plot)
```



e. Interpret your figure

Differences in copepod and ostracod abundance vary across sites, with South of Venoco Bridge having the highest average copepod abundance with Devereux Creek following. This contrasts average ostracod abundance as South of Venoco Bridge has a very low average ostracod abundance, Devereux Creek has a similar amount of ostracods, and overall South of Phelps Bridge has the highest average ostracod abundance.

Ostracod abundance seems to not follow site differences in salinity / it appears to be influenced by other factors. For example, South of Phelps Bridge and Devereux Creek has the highest ostracod abundance, and also has low salinity (ppt), however so does Phelps Road and North of Phelps Bridge however they have low ostracod abundances.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

c. Code your figure

d. Interpret your figure